

Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

Final Individual Presentation & Submission

Student: C. Nitesh Kumar

Day-28: Final report

Final Project Status

Technical Work: Completed

Mentor Review: Approved

Final Package: Ready for Submission

What Was Developed

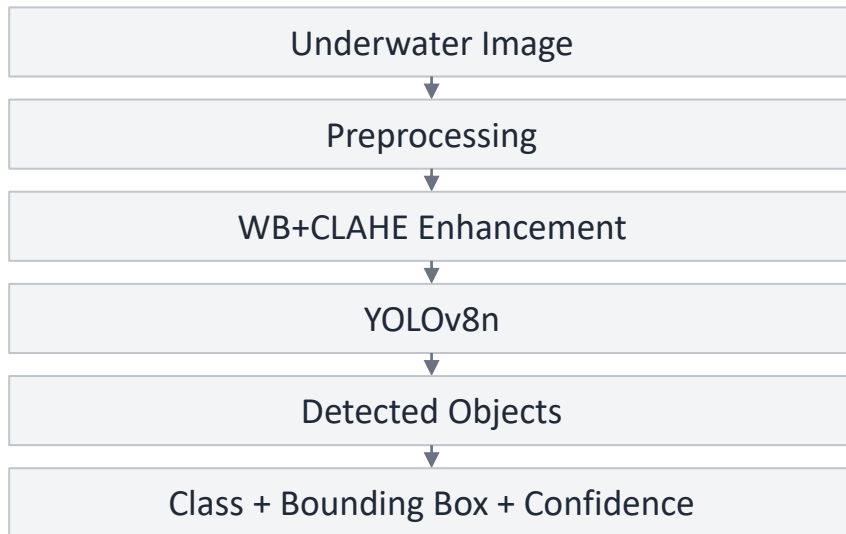
Final System

An end-to-end underwater object detection pipeline:

- Dataset inspection
- Annotation validation
- Image preprocessing
- Underwater image enhancement
- YOLOv8n object detection
- Performance evaluation
- Runtime analysis
- Error and failure analysis

Final Output

Object Class + Bounding Box + Confidence Score



Complete Experimental Journey

Day	Work Completed
Day 1	Problem understanding & pipeline
Day 2	Literature review & methods
Day 3	Research framing & clarification
Day 4	Dataset schema & risk analysis
Day 5	Experimental design
Day 6	Methodology definition
Day 7–12	Dataset audit, EDA & preprocessing
Day 13–15	WB+CLAHE & enhancement analysis
Day 16	Original YOLOv8n baseline

Day	Work Completed
Day 17–18	Enhanced detection experiments
Day 19	Hyperparameter tuning
Day 20	Robustness & ablation analysis
Day 21–22	Candidate review & model freeze
Day 23	Final validation & test evaluation
Day 25	Error/failure analysis
Day 26	Final software package
Day 27	Mentor review & sign-off
Day 28	Peer presentation & submission

Key Point

The project progressed from problem understanding to a reproducible final detection pipeline.

Final Experimental Comparison: Original vs WB+CLAHE

Dataset	Configuration	Precision	Recall	mAP50	mAP50-95
DIAT	Original	90.02%	92.09%	92.72%	65.90%
DIAT	WB+CLAHE	87.70%	90.20%	91.10%	64.00%
Aquatic Plant	Original	94.64%	93.52%	98.05%	70.79%
Aquatic Plant	WB+CLAHE	94.60%	97.90%	98.70%	70.70%
Well	Original	35.60%	36.80%	36.50%	16.40%
Well	WB+CLAHE	36.30%	37.70%	35.20%	15.70%

Interpretation

- DIAT: performance decreased after enhancement
- Aquatic Plant: recall and mAP@0.5 improved slightly; mAP@0.5:0.95 almost unchanged
- Well: precision and recall increased slightly, but mAP decreased

Main Observation: WB+CLAHE does not guarantee improved object detection.

Hyperparameter Tuning & Final Selection

Aquatic Plant Tuning

Experiment	Learning Rate	Batch	Precision	Recall	mAP50	mAP50-95
H01	0.001	16	94.5%	95.4%	98.2%	71.3%
H02	0.0005	16	94.5%	94.9%	98.2%	71.4%
H03	0.0005	32	95.4%	96.1%	98.5%	71.5%

Well Tuning

Experiment	Learning Rate	Batch	Precision	Recall	mAP50	mAP50-95
H04	0.001	16	35.7%	40.1%	36.8%	16.8%
H05	0.0005	16	43.6%	33.5%	37.2%	16.4%
H06	0.001	32	34.4%	34.5%	35.3%	16.1%

Selection Rule

Primary validation metric: mAP@0.5:0.95. Selected: Aquatic Plant → H03, Well → H04.

Important Finding: validation-based selection did not always guarantee better test-set performance.

Final Test Evaluation

Aquatic Plant

Metric	H03 Final	Original Baseline	Change
Precision	97.99%	95.69%	+2.30 pp
Recall	90.26%	94.44%	-4.18 pp
mAP@0.5	96.75%	98.55%	-1.80 pp
mAP@0.5:0.95	71.12%	70.63%	+0.49 pp

Well

Metric	H04 Final	Original Baseline	Change
Precision	43.00%	44.64%	-1.64 pp
Recall	37.46%	38.31%	-0.85 pp
mAP@0.5	39.34%	42.20%	-2.86 pp
mAP@0.5:0.95	16.15%	18.46%	-2.31 pp

Final Interpretation

Aquatic Plant: mixed test-set changes.

Well: below the original baseline on all four test metrics.

Final System Performance

Runtime & Model Size

Dataset	Inference Time/Image	FPS	Model Size
DIAT	22.14 ms	45.16	5.95 MB
Aquatic Plant	9.80 ms	102.02	5.96 MB
Well	9.98 ms	100.21	5.96 MB

What This Shows

- YOLOv8n models are relatively small
- Aquatic Plant and Well exceed 100 FPS in the measured setup
- DIAT was slower due to the evaluated configuration/input characteristics
- Runtime covers YOLO inference only
- WB+CLAHE preprocessing time was measured separately

Final Model Files

DIAT_Baseline/best.pt

Aquatic_Plant_H03/best.pt

Well_H04/best.pt

Error Analysis & What We Learned

DIAT

- Fish/Fishes confusion
- Possible missed Camera detection
- Low-confidence predictions

Aquatic Plant

- Missed plant detections
- Low-confidence predictions
- Unclear plant boundaries

Well

- Small objects
- Edge-located objects
- Difficult stone detections
- Possible false positives

Overall Failure Conditions

Failure Condition	Observed In
Haze / low visibility	All datasets
Small objects	Mainly Well
Similar classes	Mainly DIAT
Class imbalance	Mainly Well
Low-confidence cases	All datasets

Learning: performance depends not only on enhancement, but also on object size, visibility, class similarity and dataset characteristics.

Final Reproducibility & Submission Package

Repository Structure

```
Underwater-vision-surveillance/  
├── Day1/ ... Day27/  
└── Day26_Final_Project/  
    ├── preprocessing/  
    ├── enhancement/  
    ├── detection/  
    ├── evaluation/  
    ├── models/  
    ├── results/  
    ├── figures/  
    ├── docs/  
    ├── README.md  
    └── requirements.txt
```

Reproducibility Checklist

Item	Status
Source code	✓ Verified
Preprocessing configuration	✓ Verified
Enhancement method	✓ Verified
YOLO configuration	✓ Verified
Final model references	✓ Verified
Evaluation results	✓ Verified
Runtime results	✓ Verified
Error-analysis outputs	✓ Verified
README	✓ Verified
requirements.txt	✓ Verified
Documentation	✓ Verified

GitHub:<https://github.com/Nitesh71332/Underwater-vision-surveillance>

Final Submission & Future Scope

Final Submission Checklist

Deliverable	Status
Final Report	✓ Completed
Final PPT	✓ Completed
GitHub Repository	✓ Verified
Final Models	✓ Available
Source Code	✓ Available
Results	✓ Available
Prediction Outputs	✓ Available
Error Analysis	✓ Available
Documentation	✓ Completed
Future Scope	✓ Included

Day 28 Status

Peer presentation, timing check, repository and reproducibility checks, final submission: Completed ✓

Future Scope

- Compare Water-Net and FUnIE-GAN
- Handle severe class imbalance
- Improve small-object detection
- Source-independent dataset splits
- Evaluate additional underwater datasets
- Add video-based object tracking
- Explore advanced detection architectures

Final Conclusion

The project built a complete underwater vision pipeline and showed that enhancement affects detection differently depending on dataset and object characteristics.